
Reproducing DiT and Probing the Geometry of the Diffusion Latent Space with Hyperbolic Space

Changhyeon Leem

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Dept. of Materials Science and Engineering
Korea University
leemcado@gmail.com

Valerie Marie Kinson

2026951550

University of Tennessee, Knoxville
Exchange Student, Korea University
vkinson1@vols.utk.edu

Abstract

We reproduce the Diffusion Transformer (DiT) of Peebles and Xie (2023), which replaces the U-Net backbone of latent diffusion with a transformer, and we verify its central claim: a scaling law relating compute (GFLOPs) to generation quality (FID). Across both the model-size and patch-size axes our re-implementation recovers the law, with a log-GFLOPs-to-FID correlation of $r = -0.992$, even cleaner than the -0.93 reported in the original paper. Building on this reproduction, we form and test a hypothesis about the geometry of the diffusion latent space, which we call HiT (Hyperbolic DiT): that a well-trained DiT encodes a radial, hierarchical structure in its predicted clean image $\hat{x}_0$, which we attempt to expose and control by lifting the latent into hyperbolic space. We find that standard DiT’s $\hat{x}_0$ does draw a previously-unreported radial, U-shaped structure consistent across three model scales; however, a causal-separation experiment shows the FID improvement from our sample-wise loss comes from added global information, not from hyperbolic geometry, and directly forcing the structure degrades performance. We explain the U-shape analytically through a Tweedie decomposition, arguing that the late-timestep divergence is a posterior phenomenon—a trace of denoiser error—which is precisely why it cannot be controlled into a performance gain. Code is available at https://github.com/leemcado/AML_26.

1 Introduction and Motivation

Before DiT, the standard backbone for diffusion models was the U-Net (Ronneberger et al., 2015). U-Net is a CNN-based model whose inductive bias for capturing multi-scale structure is baked directly into the network architecture. The catch is that obtaining this benefit requires a fixed network shape: U-Net-based diffusion was powerful, but its architecture was rigid. The Transformer (Vaswani et al., 2017) instead learns relations between tokens through query–key–value attention, placing its inductive bias at the layer level rather than the architecture level, and it has become a general-purpose standard across many domains. In particular, the Vision Transformer (ViT) (Dosovitskiy et al., 2021) demonstrated a scaling law in vision, showing that transformer layers transfer into the visual domain.

In this context, DiT (Peebles and Xie, 2023) replaced the diffusion backbone, swapping the U-Net for a transformer. We reproduce DiT ourselves and verify its key claim, the scaling law. While reproducing DiT, we further formed a hypothesis about the diffusion latent space and tested it directly. This second project is HiT, for Hyperbolic DiT, since we reinterpret the DiT latent space with hyperbolic geometry. We present it honestly, in the order in which we actually reasoned—leaps included—rather than as a story polished to fit the results.

2 Contributions

- We faithfully reproduce DiT and verify its FLOPs scaling law along both axes (model parameters and patch size), obtaining a log-GFLOPs-to-FID correlation of -0.992 , cleaner than the paper’s -0.93 .
- We identify a previously-unreported radial, *U-shaped* geometric structure in standard DiT’s $\hat{x}_0$ latent, consistent across the S/B/L scales.
- We design a causal-separation experiment that isolates the source of an observed FID gain, showing it comes from adding sample-wise global information rather than from hyperbolic geometry.
- We give an analytic (Tweedie / posterior-mean) explanation of the U-shape, framing the late-timestep divergence as a posterior artifact of denoiser error, which explains why direct control of the structure fails.

3 Method

3.1 DiT (reproduced)

DiT trains and generates not on pixels but on the latent that a pretrained VAE compresses. A DiT block is self-attention plus adaptive layer normalization (AdaLN) for conditioning; the paper gave three conditioning options and we adopt AdaLN, which is known to work best. The model output predicts the added noise together with a covariance, so the loss combines a noise-prediction term with the KL term of the reverse distribution (Ho et al., 2020; Nichol and Dhariwal, 2021):

$$L_{simple} = \mathbb{E}_{t,x_0,\epsilon} [\|\epsilon - \epsilon_\theta(x_t, t)\|^2], \quad (1)$$

$$L_{vlb} = \mathbb{E}_{t,x_0,\epsilon} [D_{KL}(q(x_{t-1} | x_t, x_0) \| p_\theta(x_{t-1} | x_t))], \quad (2)$$

$$L_{total} = L_{simple} + \lambda L_{vlb}. \quad (3)$$

Because we assume Gaussians, the covariance alone suffices to compute the KL. The attention block implements the standard multi-head formulation,

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V, \quad (4)$$

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O, \quad (5)$$

and we add a 2D sinusoidal positional embedding to retain the locality inductive bias of vision, with a 1D embedding used for the timestep. AdaLN acts globally during training, so if the variance grows too large, training can collapse; we therefore initialize every weight and bias of the AdaLN modulation block to zero, making each block an identity function at the start (adaLN-Zero). Crucially, learning still happens: by the chain rule the gradient on these weights is nonzero,

$$\frac{\partial L}{\partial W} = \frac{\partial L}{\partial \alpha} \cdot \frac{\partial \alpha}{\partial W} = \frac{\partial L}{\partial \alpha} \cdot c^\top, \quad (6)$$

so a block starts as an identity yet stays trainable, which aids stable convergence. The final output is a $(B, 8, 32, 32)$ tensor: four noise channels and four covariance channels.

For evaluation we generate with classifier-free guidance (CFG): since AdaLN enables class-conditional generation, we can sample specific classes without a separate discriminator. To enable CFG we drop the class token to a null token 10% of the time during training, so the model also learns unconditional generation,

$$\hat{\epsilon}_\theta(x_t, c) = \epsilon_\theta(x_t, \emptyset) + w(\epsilon_\theta(x_t, c) - \epsilon_\theta(x_t, \emptyset)), \quad (7)$$

with guidance scale $w = 4$ by default. Generation is reverse sampling, each step computing $x_{t-1} = \mu_\theta(x_t, t) + \sigma_\theta z$ and masking the noise at $t = 0$. Training uses all 1000 steps, while sampling is accelerated to 250 steps with SpacedDiffusion under the same noise schedule; the reconstructed latent is then decoded back to image space with the VAE. Our codebase separates configuration, diffusion (forward/reverse process), models (the PyTorch architecture), and evaluation (FID).

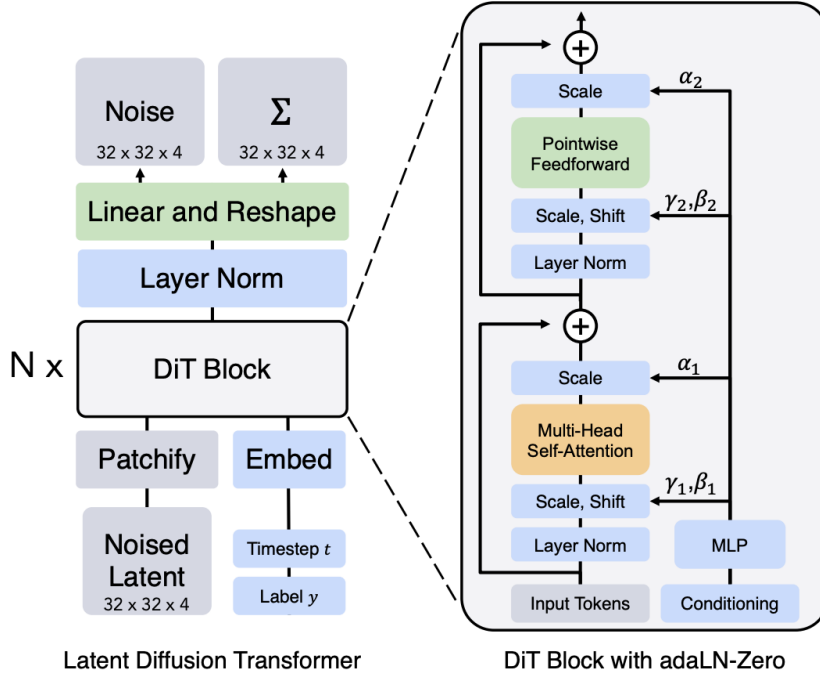


Figure 1: The DiT architecture: a latent diffusion model with a transformer backbone and an adaLN-Zero block for conditioning. Adapted from Peebles and Xie (2023).

3.2 HiT (ours)

Motivation and hypothesis. Diffusion does not only learn clean data; it uses noisy data across all timesteps, and all noise levels are handled by one shared, time-conditioned network rather than a separate network per t . Since time is continuous, we reasoned that this space encodes everything from data to full noise as one continuous structure—consistent with score-based models that learn a continuous family of noise-smoothed distributions with a single model (Song and Ermon, 2019; Song et al., 2021). Moreover, any data taken to full noise converges to the same Gaussian, so all data forms one structure joined at a common terminal, and generation can be read as a weight-tied recurrent system applied many times in the latent. This view is supported by the interpretation of diffusion as spectral autoregression (Dieleman, 2024) and by analyses of looped, weight-tied models converging to consistent trajectories and stable fixed points (Blayney et al., 2026). From these intuitions we made a guess: a well-trained diffusion would have a *radial* manifold, with the common terminal (noise) at the center and specific data spreading outward (Figure 2). We want models that extract a latent hierarchy with a suitable inductive bias; if the manifold is radial, the noise that commonly generates different classes—a kind of common ancestor—would be encoded there too. We therefore reinterpret the DiT latent with hyperbolic geometry, which holds hierarchy best. We acknowledge up front that this argument contains a reversed-causality leap, which we knew and ran anyway.

Preliminary. Hyperbolic space has negative curvature, so volume grows exponentially with distance from the origin, matching a tree whose node count grows exponentially with depth—hence a “continuous tree.” When embedding, abstract roots sit near the origin and specific leaves near the boundary, and distance to the origin is reported to measure generality or uncertainty (Khruikov et al., 2020); this fits the intuition that the center is noise and the outside is data. As a precedent for controlling a model through its latent, the LeWorldModel of LeCun et al. (2026) showed that controlling the last-layer latent controls the whole model’s dynamics; with the same idea, our analysis target is $\hat{x}_0$, reconstructed from the noise prediction—the clean image the model currently has in mind.

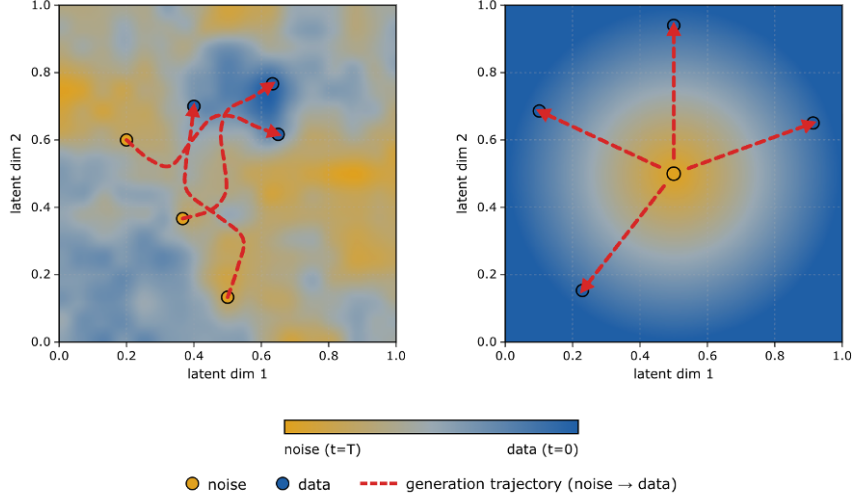


Figure 2: The radial-manifold hypothesis. *Left*: in a rough, poorly-shaped latent landscape, generation trajectories from noise to data wander without clear organization. *Right*: the idealized *radial* manifold we hypothesize for a well-trained diffusion—the common terminal (noise, $t=T$) sits at the center and specific data ($t=0$) spread outward, so each trajectory runs radially from center to a data point. This is the structure we attempt to expose and control with hyperbolic geometry.

Pipeline and losses. From the ϵ prediction we reconstruct $\hat{x}_0$, project it from 4096 to 256 dimensions, and lift it onto the Poincaré ball with the exponential map (Figure 3). The projection head is constrained to be orthogonal (a Stiefel-manifold constraint, restored each step by a QR retraction) to prevent the trivial zero-norm solution. The hyperbolic sample-wise loss measures how close the predicted sample is to the target in hyperbolic space:

$$d(z, z^*) = \frac{2}{\sqrt{c}} \operatorname{artanh}(\sqrt{c} \|\ominus z^* \oplus z\|), \quad L_{hyp} = \mathbb{E}[d(z, z^*)^2], \quad (8)$$

which nudges the latent to follow hyperbolic geometry (Khrulkov et al., 2020). A second, stronger regularizer enforces the hypothesized geometry directly, pulling the origin-distance to decrease linearly with t toward a target of $2(1 - t/T)$:

$$L_{reg} = \mathbb{E}\left[\left(d^0(z) - 2\left(1 - \frac{t}{T}\right)\right)^2\right]. \quad (9)$$

We measure the structure two ways: qualitatively, with a Poincaré UMAP of the 256-dim latents projected to 2D; and quantitatively, with how each origin-distance changes with t , using Spearman’s ρ to test for a monotonic decrease.

4 Experiments

4.1 Reproducing the DiT scaling law

What the paper presents as DiT’s core contribution is a scaling law over FLOPs, which can be controlled two ways: the number of model parameters and the patch size. A larger patch makes the token sequence shorter, lowering total FLOPs. We tested whether this scaling law actually reproduces. For model size we used the same hyperparameters as the paper (Table 1), with CFG scale 4, learning rate 10^{-4} , and batch size 256. Due to resource limits we trained only to 100K steps and excluded XL; training ran on four A100 GPUs, parallelized with DDP. For patch size we tested the paper’s options 2, 4, 8: doubling the patch quarters the token count, giving DiT-S/2, S/4, S/8 about 1.4, 0.4, 0.1 GFLOPs respectively.

Results. On a log scale, larger models converged faster, and FID dropped as scale grew, so the model-size scaling law holds quantitatively (Table 2); sample quality also rises with model size

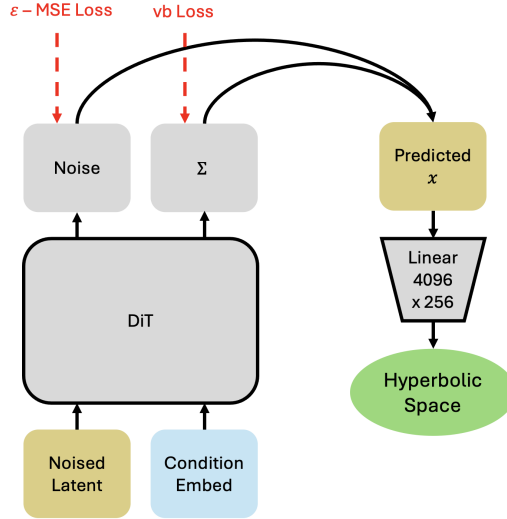


Figure 3: The HiT pipeline. From the DiT output we reconstruct $\hat{x}_0$, project 4096 $\rightarrow$ 256 dimensions, and lift it onto hyperbolic space.

Table 1: Model-size settings used for the parameter scaling study (after Peebles and Xie, 2023).

Model	Layers N	Hidden size d	Heads	GFLOPs ($I=32, p=4$)
DiT-S	12	384	6	1.4
DiT-B	12	768	12	5.6
DiT-L	24	1024	16	19.7
DiT-XL	28	1152	16	29.1

Table 2: Model-size scaling (patch 2, 100K steps).

Model	#params	step/s	loss	FID $\downarrow$
DiT-S/2	33M	6.7	0.156	61.50
DiT-B/2	130M	2.8	0.153	35.90
DiT-L/2	458M	0.9	0.150	14.71

Table 3: Patch-size scaling (DiT-S, 100K steps).

Model	patch	step/s	loss	FID $\downarrow$
DiT-S/2	2	6.7	0.156	61.50
DiT-S/4	4	17.8	0.164	109.19
DiT-S/8	8	18.4	0.175	115.05

qualitatively. Likewise, as FLOPs grow—that is, as the patch size shrinks—convergence gets faster and FID improves (Table 3). Relating compute to quality directly, the log of GFLOPs and FID show an almost perfect negative correlation, with a coefficient of -0.992 (Figure 4). That is a cleaner trend than the paper’s -0.93 , so we conclude that the scaling law claimed in the paper is reproducible.

4.2 HiT experiments

We designed the experiments in three stages, by how strongly we intervene. **(1) Observation:** we attach only a frozen analysis head to an untouched standard DiT at three scales and check whether the radial structure exists—the plain DiT from the paper, with no regularization, analyzing the full 4096 dimensions. **(2) Light intervention:** we add the hyperbolic sample-wise MSE to the DiT loss as a hybrid, and we add a control that drops the hyperbolic part and passes the same sample-wise loss through Euclidean space, to test whether any change comes from the hyperbolic geometry. **(3) Strong intervention:** we enforce the structure directly with the regularizer, in variants over the range $t \leq 800$ versus all t , and measured relative to the origin versus the manifold mean, including a variant pushing opposite to the intuition.

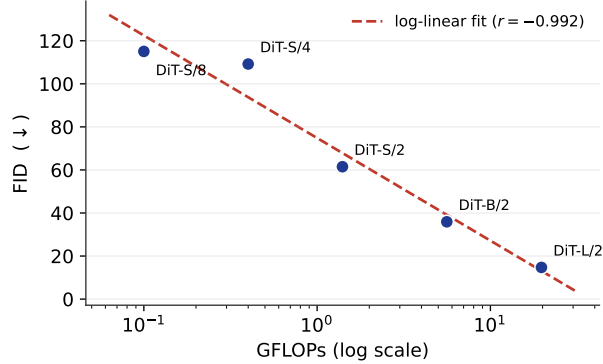


Figure 4: Compute versus generation quality for our DiT runs: log-GFLOPs versus FID, with a log-linear fit ($r = -0.992$).

Table 4: Hybrid loss and the causal-separation control (DiT-B/2, 130M).

Model	loss	steps	FID ↓
DiT-B	standard	100K	35.90
HiT-B	+ hyperbolic MSE	100K	21.89
HiT-B	+ hyperbolic MSE	60K	44.52
DiT-B	+ Euclidean (control)	60K	35.50

Table 5: Regularization (HiT-B/2, 60K). ρ is the frozen Spearman correlation.

Target	t	FID ↓	ρ
manifold mean	800	52.54	-0.08
manifold mean	1000	85.29	-0.46
origin	800	50.55	-0.05
origin	1000	90.92	-0.55
mean (rev.)	800	44.48	+0.56
mean (rev.)	1000	45.08	+0.65

Results. The first result was the most surprising: the radial structure really exists. At all three scales, t and origin-distance show a consistent negative correlation (Spearman $\rho = -0.59, -0.64, -0.44$), and UMAP shows higher-noise points gathering toward the center (Figure 5). But one thing broke the intuition: from $t = 0$ to about 800 the origin-distance drops monotonically, yet from 800 to 1000 it diverges, giving a U shape—and this emerged in a plain DiT with no regularization, which we read as a positive sign. The hybrid loss improved FID markedly, from 35.9 to 21.9 (Table 4). However, the Euclidean control improved it *even more*, from 44.5 to 35.5 (Table 4): so the gain came not from hyperbolic geometry, but from adding sample-wise global information to a DiT that had only learned a patch-wise local loss. The hyperbolic part actually hurts—a painful result for the hypothesis, but a clean attribution. Finally, forcing the structure with the regularizer consistently hurt more the harder we pushed it (Table 5, Figure 6); notably, the reversed regularizer hurt *less* than the intuition-aligned one and did not damage the monotonic descent up to 800, only accelerating the late divergence. Forcing the structure to match the intuition was the bigger loss.

5 Own Analysis and Improvement

Critical assessment of the reproduced method. DiT’s strengths are clear: a clean, quantitatively reproducible scaling law, and the architectural flexibility of placing the inductive bias at the layer level rather than baking it into a rigid backbone. From our standpoint its main unexplored weakness is the geometry of what the model represents in its latent space—which is exactly what motivated HiT.

Interpreting the negative result. We designed this project to form a hypothesis about the diffusion latent space and to create value by controlling it. We failed at the control. Instead, we find meaning in a phenomenon we could not find reported anywhere: the radial, U-shaped structure that a standard DiT’s $\hat{x}_0$ draws. So was the hypothesis right? Honestly, we are no longer sure: the hypothesis was a reversed-causality leap, and hyperbolic space was not the best tool. Moreover, the structure can be explained by a formula. In the forward process $x_t = \sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \epsilon$, the reconstruction

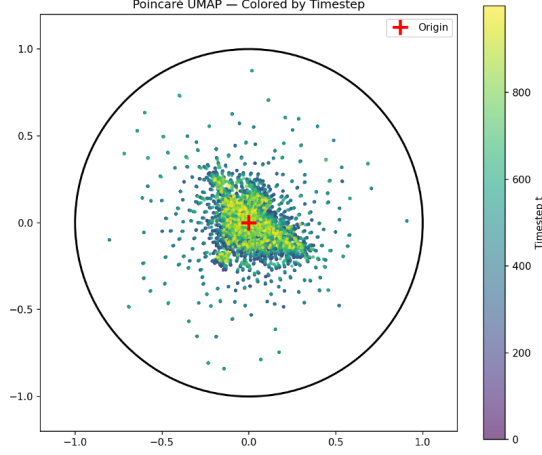


Figure 5: The observed latent structure. A Poincaré UMAP of a standard DiT’s $\hat{x}_0$ latents, colored by timestep t : high-noise points (bright, large t) concentrate near the center (the origin, red +) while lower-noise points spread outward—the radial organization our hypothesis predicted. Across scales, origin-distance and t are negatively correlated (Spearman $\rho = -0.59, -0.64, -0.44$).

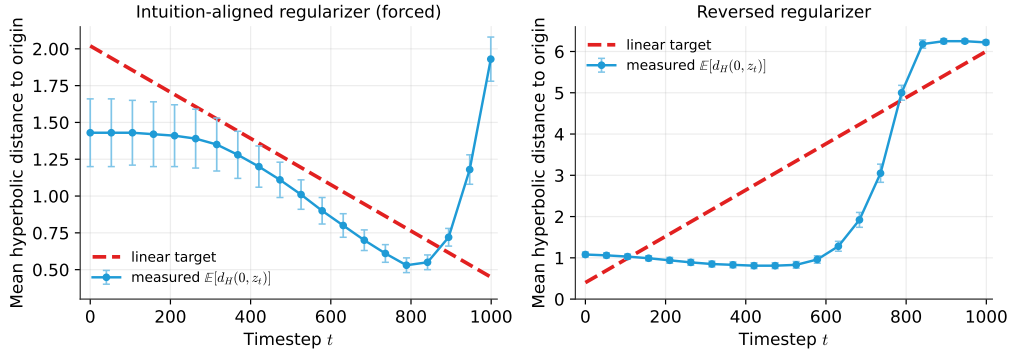


Figure 6: Strong intervention (regularization): measured $d_H(0, z_t)$ versus t under two linear regularizers; the red dashed line is the regularizer’s linear *target*. *Left*: the intuition-aligned regularizer forces the origin-distance down toward the decreasing target, yet FID degrades. *Right*: the reversed regularizer pushes the target upward—the monotonic descent up to $t \approx 800$ is preserved and only the late divergence accelerates, and it hurts *less* than the intuition-aligned one (Table 5).

is $\hat{x}_0 = (x_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta) / \sqrt{\bar{\alpha}_t}$. Splitting the noise prediction into an expectation and an error, $\epsilon_\theta = \mathbb{E}[\epsilon | x_t] + \delta$, and substituting gives

$$\hat{x}_0 = \mathbb{E}[x_0 | x_t] - k_t \delta, \quad k_t = \frac{\sqrt{1 - \bar{\alpha}_t}}{\sqrt{\bar{\alpha}_t}}. \quad (10)$$

The first term—the posterior mean, or Tweedie estimate—shrinks toward the data mean as t grows, pulling the origin-distance down: this is the monotonic-descent term. The coefficient of the second term, k_t , diverges as t grows (since $\bar{\alpha}_t \rightarrow 0$), so as long as the error δ is nonzero it pushes the origin-distance back up at large t : this is the rising term. But if the diffusion were perfectly trained and matched the distribution, then $\delta = 0$, the noise-driven rising term is suppressed, and only a pure monotonic descent remains. In other words, the divergence after 800 is not the essence of the structure but a trace of model error—a *posterior* phenomenon that follows from training (Figure 7).

This is why the attempt to control it directly and turn it into performance failed: one cannot turn a posterior artifact into an inductive bias. It is like banning ice-cream sales to reduce shark attacks; both are effects of a common cause—here, the quality of the denoiser. Even so, this was a genuinely

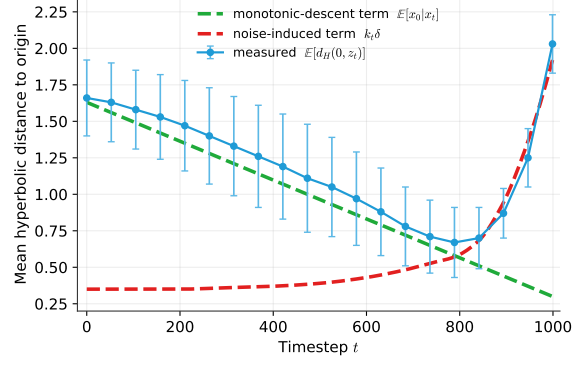


Figure 7: The radial U-shape, explained. The measured mean origin-distance (blue) decomposes into a monotonic-descent term from the posterior mean $\mathbb{E}[x_0 | x_t]$ (green) and a noise-induced term $k_t \delta$ from the prediction error (red). The green term drives the descent toward the center; the red term, whose coefficient $k_t \rightarrow \infty$ as $t \rightarrow T$, drives the late-timestep rise. The U-shape is therefore a posterior artifact of nonzero error δ , and it vanishes in the limit of a perfectly trained model.

valuable first experience of forming a hypothesis, designing the experiments, running them, and judging the results.

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